

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=butter

retailer=silpo_novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

	series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
	producer	0.08225840900720094	0.01	2.6818026212664985e-12	0.1	1.3006957963949979e-20	0.1	I(1)	1.0	butter	silpo_novus	promo_controlled	nan	nan
	prozorro	0.05640422694264767	0.01	3.518946761497874e-22	0.1	1.9269184182015793e-13	0.1	I(1)	1.0	butter	silpo_novus	promo_controlled	nan	nan
	retail	0.009658462541496344	0.015275724253247233	0.00042417475534266833	0.1	0.0016100992365552125	0.1	I(1)	0.0	butter	silpo_novus	promo_controlled	nan	nan
	producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	butter	silpo_novus	promo_controlled	0.46279135586174147	220.0
	prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	butter	silpo_novus	promo_controlled	1.6643849009237306e-08	51.0
	producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	butter	silpo_novus	promo_controlled	0.9141905239557406	55.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
butter	silpo_novus	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.2127764229425364	0.27405320585302084	-0.8083252331134949	1.2523745815882856e-26	0.787381486847105	0.20112635504343956	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
butter	silpo_novus	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	51	0.3808399108271803	5.038120798935989	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.02219; b... Primary chain is strictly...	
butter	silpo_novus	promo_controlled	daily	prozorro_to_retail	ECM	retail	prozorro	49	-0.019317533092633524	-0.11304424771502886	-0.5842642909954743	0.007904731650635162	nan	nan	nan	ok	0	engle_granger_p=0.06; lag... Primary chain is strictly...	
butter	silpo_novus	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	49	0.03764617386887585	-0.8051638051994188	-0.9346811115345983	0.0012434236716764512	0.9262473301905156	0.32065369008869127	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
NARDL	producer_to_prozorro	prozorro	producer	0.30437404399912776	0.5287391650637225	4.2203813853019655e-84	0
ARDL	prozorro_to_retail	retail	prozorro	0.15171373599684104	0.9428759814198081	0.1216394133775761	0
ECM	prozorro_to_retail	retail	prozorro	0.7561861082620142	0.13081480015773286	1.4411688625167e-05	0
NARDL	prozorro_to_retail	retail	prozorro	0.4526018945159781	0.1773413029580844	0.0008392768825093986	0

Table: SeriesUsed

	date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
	2021-02-01	31.78268739915295	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-02	31.74153855497394	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-03	31.71205395008976	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-04	31.69228954461795	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-05	31.68030129867603	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-06	31.67414517238153	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-07	31.67187712585198	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-08	31.6715531192049	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-09	31.71501589339797	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-10	31.82710410052748	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-11	31.98036756741884	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-12	32.14735612089748	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-13	32.30061958778885	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-14	32.41270779491835	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-15	32.45617056911143	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-16	32.45341900628176	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-17	32.44522030312711	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-18	32.43165843764893	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-19	32.41281738784873	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-20	32.38878113172797	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-21	32.35963364728814	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-22	32.32545891253073	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-23	31.91148084818262	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-24	30.90284056637734	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-25	29.53949783222626	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-26	28.06141241084077	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-27	26.70854406733228	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-02-28	25.72085256681216	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-01	25.33829767439181	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-02	25.36591783597624	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-03	25.44026511447978	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-04	25.54856970052784	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-05	25.67806178474581	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-06	25.81597155775909	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-07	25.94952921019307	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-08	26.06596493267315	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-09	26.18214222460376	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-10	26.31503339825227	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-11	26.45203186353847	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...
	2021-03-12	26.58053103038216	nan	nan	butter	silpo_novus	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	0.07873337531291687	
producer_to_prozorro	2	0.08283043430904553	
producer_to_prozorro	3	0.08251800718380085	
producer_to_prozorro	4	0.08422490303011267	
producer_to_prozorro	5	0.0907848031216401	
producer_to_prozorro	6	0.08185920607839353	
producer_to_prozorro	7	0.08099913022176998	
producer_to_prozorro	8	0.08418185337829556	
producer_to_prozorro	9	0.08011971167494186	
producer_to_prozorro	10	0.08071177659888301	
producer_to_prozorro	11	0.0855520813606708	
producer_to_prozorro	12	0.08891786146062777	
producer_to_prozorro	13	0.0906935444584109	
producer_to_prozorro	14	0.09336004953876562	
producer_to_prozorro	15	0.09806722582073225	
producer_to_prozorro	16	0.10411102969445078	
producer_to_prozorro	17	0.11083781027515911	
producer_to_prozorro	18	0.11554030327110705	
producer_to_prozorro	19	0.11683243707057434	
producer_to_prozorro	20	0.11514579002515733	
producer_to_prozorro	21	0.11293989092313386	
producer_to_prozorro	22	0.11348584116046659	
producer_to_prozorro	23	0.11614446866624657	
producer_to_prozorro	24	0.1193793796780393	
producer_to_prozorro	25	0.12118388938915296	
producer_to_prozorro	26	0.11984324455041497	
producer_to_prozorro	27	0.11226973352035723	
producer_to_prozorro	28	0.1007561552537856	
producer_to_prozorro	29	0.09278063269691687	
producer_to_prozorro	30	0.09120495228803738	
prozorro_to_retail	1	-0.14268264969443448	
prozorro_to_retail	2	-0.02343528234199047	
prozorro_to_retail	3	-0.05296137558219399	
prozorro_to_retail	4	-0.30502570274713703	
prozorro_to_retail	5	-0.005989877695682057	
prozorro_to_retail	6	-0.019821237526080906	
prozorro_to_retail	7	0.03241209316341742	
prozorro_to_retail	8	-0.30293310013184976	
prozorro_to_retail	9	-0.2796119360021343	
prozorro_to_retail	10	-0.10698033472121114	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
butter	silpo_novus	promo_controlled	daily	I(1)	I(1)	I(1)	1793	230	58	220	51	0	Admissible for ARDL/ECM s...

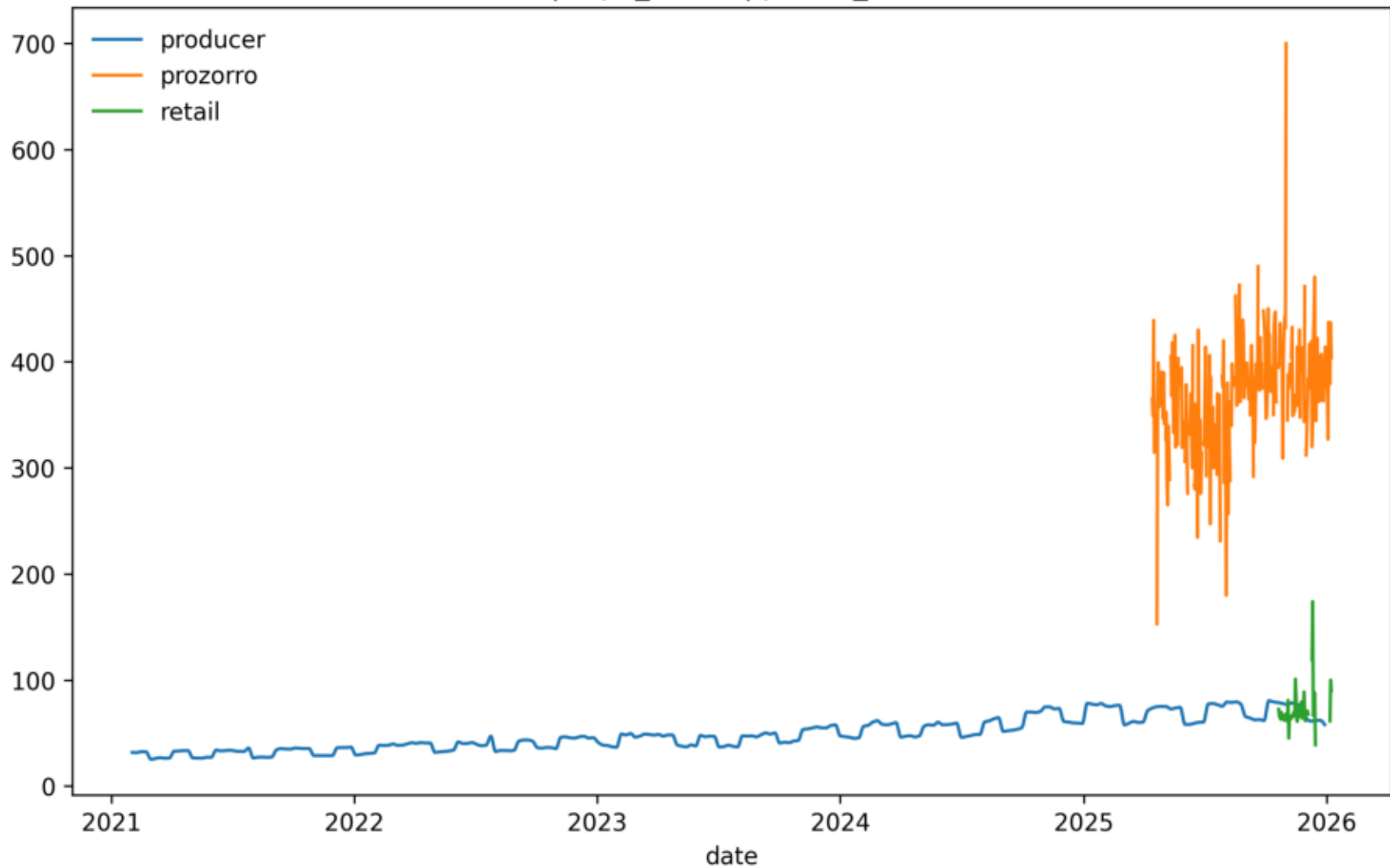
Table: NARDL_Multipliers

horizon	mult_pos	mult_neg	link
0	-0.5437076425441754	-0.33093121960163896	producer_to_prozorro
1	-0.48043463639437756	-0.2924196897746266	producer_to_prozorro
2	-0.48779791951561324	-0.29690140029865086	producer_to_prozorro
3	-0.4869410305889971	-0.2963798492381926	producer_to_prozorro
4	-0.4870407495079179	-0.29644054381166113	producer_to_prozorro
5	-0.4870291448967695	-0.29643348058903957	producer_to_prozorro
6	-0.48703049536267273	-0.2964343025589613	producer_to_prozorro
7	-0.4870303382046144	-0.2964342069036792	producer_to_prozorro
8	-0.4870303564936028	-0.2964342180353919	producer_to_prozorro
9	-0.48703035436525444	-0.2964342167399587	producer_to_prozorro
10	-0.4870303546129372	-0.2964342168907124	producer_to_prozorro
11	-0.4870303545841136	-0.29643421687316873	producer_to_prozorro
12	-0.48703035458746785	-0.2964342168752103	producer_to_prozorro
13	-0.4870303545870775	-0.29643421687497273	producer_to_prozorro
14	-0.48703035458712296	-0.2964342168750004	producer_to_prozorro
15	-0.48703035458711763	-0.29643421687499716	producer_to_prozorro
16	-0.4870303545871183	-0.29643421687499755	producer_to_prozorro
17	-0.4870303545871182	-0.2964342168749975	producer_to_prozorro
18	-0.4870303545871182	-0.2964342168749975	producer_to_prozorro
19	-0.4870303545871182	-0.2964342168749975	producer_to_prozorro
20	-0.4870303545871182	-0.2964342168749975	producer_to_prozorro
0	0.12558217957215642	0.08793600570328057	prozorro_to_retail
1	0.10968198580739312	0.07680226416172593	prozorro_to_retail
2	0.11169513896828213	0.07821192792485403	prozorro_to_retail
3	0.1114402498952836	0.07803344776904479	prozorro_to_retail
4	0.11147252187577483	0.07805604547415072	prozorro_to_retail
5	0.11146843586018941	0.07805318433659882	prozorro_to_retail
6	0.11146895319823653	0.07805354659055569	prozorro_to_retail
7	0.11146888769710238	0.07805350072490719	prozorro_to_retail
8	0.11146889599032303	0.07805350653204242	prozorro_to_retail
9	0.11146889494030292	0.0780535057967902	prozorro_to_retail
10	0.11146889507324792	0.07805350588988186	prozorro_to_retail
11	0.1114688950564155	0.07805350587809534	prozorro_to_retail
12	0.11146889505854668	0.07805350587958766	prozorro_to_retail
13	0.11146889505827685	0.07805350587939872	prozorro_to_retail
14	0.11146889505831102	0.07805350587942264	prozorro_to_retail
15	0.11146889505830669	0.07805350587941962	prozorro_to_retail
16	0.11146889505830723	0.07805350587941999	prozorro_to_retail
17	0.11146889505830718	0.07805350587941995	prozorro_to_retail
18	0.11146889505830718	0.07805350587941995	prozorro_to_retail

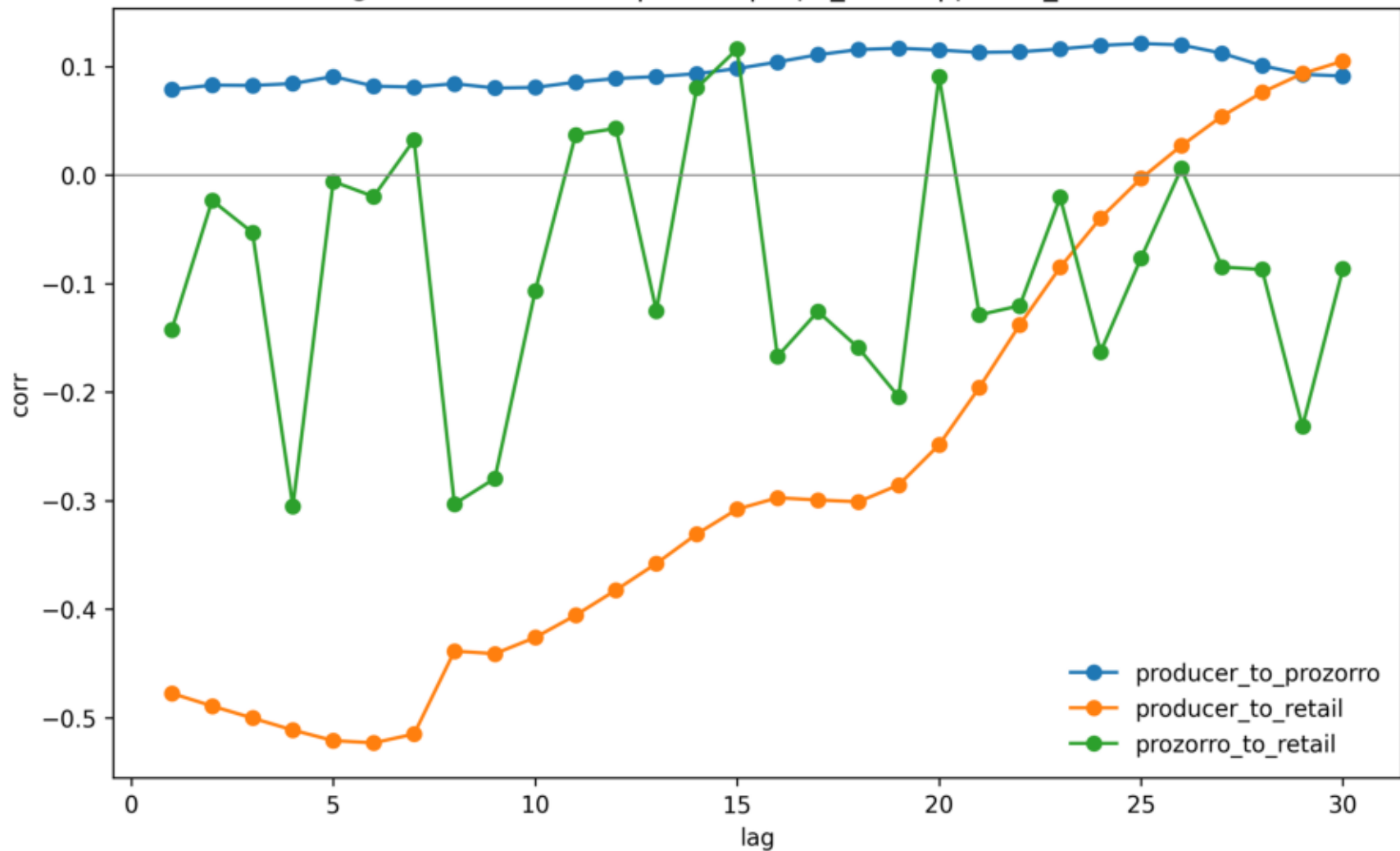
Table: VECM_IRF

No rows

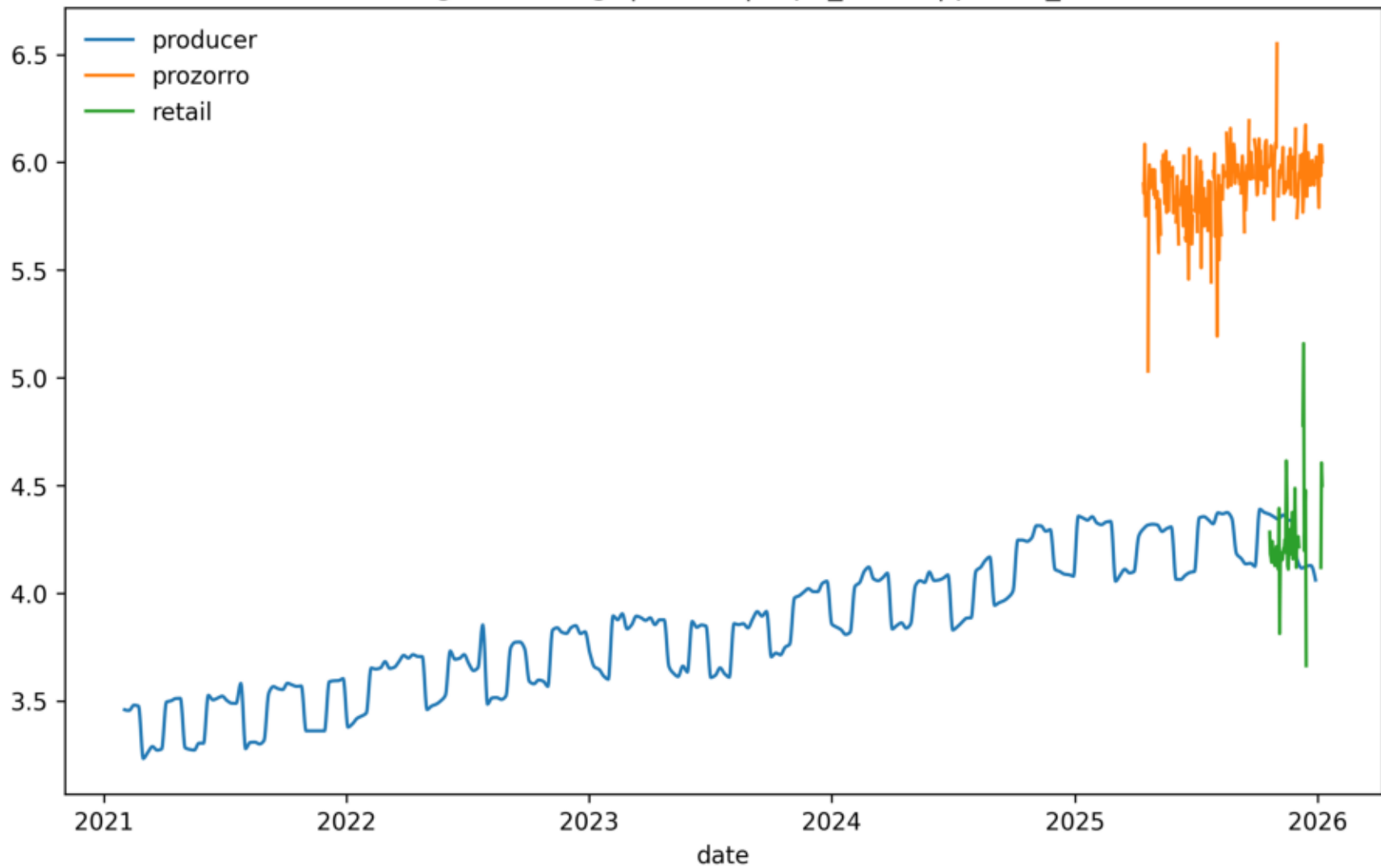
butter | silpo_novus | promo_controlled



Lag-Correlation Profile | butter | silpo_novus | promo_controlled



ECM Level Alignment (log) | butter | silpo_novus | promo_controlled



NARDL Dynamic Multipliers | butter | silpo_novus | promo_controlled

